

Do national advisories serve local consumers: an assessment of mercury in economically important North Carolina fish.

Consumption of marine fish provides both benefits (lean protein, omega-3 fatty acids and essential nutrients) and risks (main source of mercury (Hg) exposure for humans). Mercury is a potent neurotoxin and the source of more fish advisories nationwide than any other toxicant. Despite the widespread nature of Hg, it is unknown whether local Hg contamination reflects national and regional levels often used as bases to inform consumers of potential fish consumption risk. Thus, the objectives of our study were to examine Hg levels of six commonly consumed marine species harvested locally off the North Carolina coast and to compare our results to published regional (Monterey Bay Aquarium's Seafood Watch List) and national (Environmental Protection Agency, EPA, and Food and Drug Administration, FDA) Hg averages, action levels, and guidelines. We found significant differences in Hg concentrations among collected species, and we identified correlations between Hg concentration and fish length and trophic levels. Collected mahi mahi and triggerfish were below the EPA fish tissue action level (0.3ppm). Wahoo and grouper exceeded the EPA action level but were below the FDA action level (1.0ppm). King mackerel had the highest Hg concentration among targeted species, exceeding both EPA and FDA action levels. Further, our local results were not always consistent with calculated averages from EPA and FDA databases for the same species, and although many of our findings were consistent with Monterey Bay Aquarium's Seafood Watch List (southeast region), recommendations based on Hg levels would conflict with recommendations they provide based on sustainability. We find regional and national averages are not always reflective of local Hg contamination and suggest local data may be needed to accurately assess consumer risk.